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# UPMIXAI: AUTOMATIC BLIND STEREO-TO-SURROUND UPMIXING USING MUSIC SOURCE SEPARATION DEEP NEURAL NETWORKS

By

Nicholas Neng Jun Tong

A MASTERS RESEARCH PROJECT

Submitted to the Faculty
of the University of Miami
in partial fulfillment of the requirements
for the degree of Master of Science

Coral Gables, Florida

May 2025

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UNIVERSITY OF MIAMI

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# UPMIXAI: AUTOMATIC BLIND STEREO-TO-SURROUND UPMIXING USING MUSIC SOURCE SEPARATION DEEP NEURAL NETWORKS

Nicholas Neng Jun Tong

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UpmixAI: Automatic Blind Stereo-to-Surround Upmixing Using Music Source Separation Deep Neural Networks (May 2025)

Abstract of a thesis at the University of Miami.

Thesis supervised by Dr. Tom Collins
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Blind stereo-to-surround upmixing aims to convert stereo audio into a multichannel surround mix without access to original multitrack recordings. Traditional upmixing approaches rely on Primary-Ambient Extraction (PAE) using digital signal processing (DSP) techniques, which often suffer from limited generalization and spatial artifacts. This work proposes a deep learning-based alternative, integrating high-complexity state-of-the-art (SOTA) music source separation (MSS) models to enhance ambient signal extraction and stereo decorrelation.

We evaluate the efficacy of SOTA MSS architectures for separating primary and ambient components for improved spatial and spectral coherence. The selected deep neural network (DNN) architecture is trained on a custom dataset constructed from open-source music data and assessed using both objective SDR and SI-SDR metrics and a controlled participant listening study.

Experimental results from objective metrics and a controlled listening study show that MSS models can be effectively adapted for ambient signal extraction, outperforming prior deep learning approaches to the separation task and matching traditional DSP-based upmixing systems. These findings validate our proposed system as a viable and perceptually effective solution for automatic blind stereo-to-surround upmixing that utilizes deep learning throughout the entire processing chain.

*I dedicate this work to my family, friends, colleagues, and mentors, whose support and encouragement have guided me throughout my research and academic journey.*

# Acknowledgments

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Nicholas Neng Jun Tong

*University of Miami*

*May 2025*

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# CHAPTER 1

# Introduction

## 1.1 Context and Motivations

The evolution of audio consumption from mono to stereo, and now to surround sound, reflects a continuous pursuit of increasingly immersive and realistic listening experiences [12]. Despite these advancements, a significant gap remains in making high-quality surround sound content as accessible and widespread as stereo audio.

The typical workflow for a mixing engineer to create surround sound content requires access to the isolated stems of a song in order to effectively produce a multichannel mix in formats such as 5.1 or 7.1. However, in scenarios where multichannel output is desired but only the stereo mix is available, a blind upmix of the stereo content becomes necessary. Without access to the original multitrack recordings or stems, the process of blind audio upmixing from stereo to multichannel surround remains a significant challenge. Traditional approaches to upmixing have included manual mixing to a multichannel format, surround matrix decoding [11], and algorithmic digital signal processing (DSP) techniques that aim to decompose the stereo signal into separate components based on interchannel correlation [13–16].

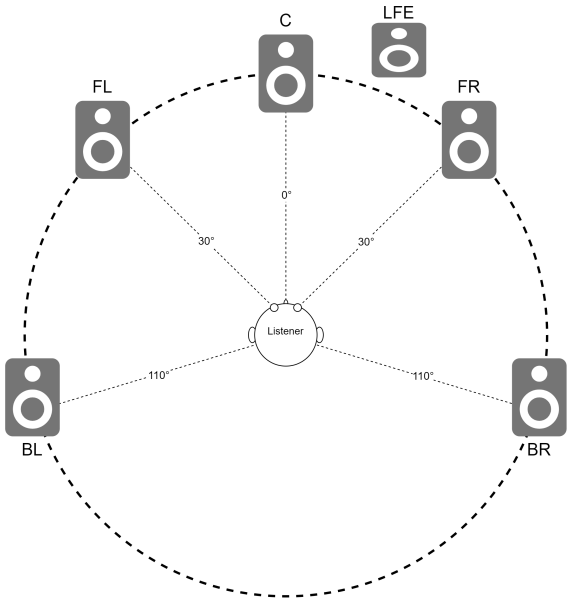



Figure 1.1: A 5.1 surround sound speaker array.

## 1.2 Problem Identification and Statement

Among the various approaches to upmixing, one of the most common in blind scenarios is Primary-Ambient Extraction (PAE), which focuses on decomposing stereo audio into primary and ambient signal components by analyzing time and frequency correlations between the stereo channels. The processed output is then spatially distributed across the surround channels. While effective, algorithmic methods such as those proposed in [11, 13–16] still exhibit several limitations:

- **Lack of Generalization:** Traditional methods rely on predefined assumptions about signal characteristics, making them less adaptable to diverse musical content.

- **Limited Context Awareness and Artifacts:** Processing audio in short, independent frames without considering long-term contextual dependencies often leads to incoherence and audible artifacts. This includes spectral and phase inconsistencies, as well as other distortions that manifest as perceptual discontinuities in the output.

By framing the upmixing problem as a separation task, deep neural networks (DNNs) offer a wide range of potential solutions by leveraging advanced techniques such as autoencoder-based latent space representations, attention mechanisms, and hybrid time-frequency processing. DNNs have already demonstrated strong efficacy in the field of music source separation (MSS), successfully isolating musical stems with high accuracy while preserving temporal and spectral coherence [1,3,9,17–29]. These innovations enable models to efficiently extract complex audio features and capture long-term dependencies, making them well-suited to learn the intricate characteristics of decorrelated ambient signal components from stereo audio.

## 1.3 Research Question and Hypotheses

While DNNs have been successfully applied in MSS and other complex audio processing tasks, their potential for direct ambient signal extraction in the context of surround upmixing remains largely unexplored. Existing DNN-based upmixing approaches still exhibit several key limitations:

- **Lack of Research Focus:** There is a notable gap in research and published work specifically addressing DNN-based PAE for upmixing. Most existing solu-

tions continue to rely on traditional DSP techniques, with the majority of DNN applications focused on instrument and dialogue separation tasks [7, 8].

- **Simplistic Neural Architectures:** Existing DNN-based upmixing approaches primarily employ basic feature extraction pipelines and shallow network designs, lacking the advanced separation capabilities demonstrated by state-of-the-art MSS models [10, 30, 31].

Thus, the fundamental research question guiding this thesis is: How can advanced MSS architectures be adapted for ambient signal extraction to enable a more effective blind stereo-to-surround upmixing system?

## 1.4 Structure of the Thesis

The remainder of this thesis is organized as follows:

- **Chapter 2: Literature Review** – Provides an overview of MSS techniques and stereo-to-surround upmixing approaches, covering both classical DSP-based methods and recent developments in deep learning. Key architectures, datasets, and integration strategies relevant to PAE are reviewed in depth.

- **Chapter 3: Methodology** – Describes the proposed solution, research motivation, and experimental design. This includes system architecture design, dataset procurement and preparation (PAEDB), model selection and training strategy.

- **Chapter 4: Evaluation and Performance Metrics** – Details the setup and execution of the evaluation process, including both the subjective listening study

and objective metric assessments. Participant demographics, stimuli selection, study design and procedures, and metric computation methods are described.

- **Chapter 5: Results** – Presents and analyzes the experimental results. This includes model training curves, computational resource comparison, statistical analysis of subjective MUSHRA listening scores across multiple perceptual dimensions, and objective performance results based on SDR, SI-SDR, and LSD metrics.

- **Chapter 6: Conclusion** – Summarizes the findings and contributions of the thesis, discusses the implications of the results, acknowledges limitations in data and methodology, and proposes directions for future research and improvement.

# CHAPTER 2

# Literature Review

## 2.1 Introduction to Music Source Separation

Music Source Separation (MSS) is the process of decomposing a mixed audio signal into its individual components, often referred to as "stems." In most music separation tasks, these stem sources typically include primary musical components such as vocals, bass, drums, and other instrumental sources [27]. MSS has a broad range of applications, from rearranging and remixing the original source mixture [32], speech enhancement [33], vocal removal for instrumental tracks [20], and most relevant to the thesis, audio upmixing [7,8]. These scenarios share a similar need for modifying or removing specific elements from the target mixture track, which historically has proven to be a challenging task due to the complex nature of overlapping frequencies and highly correlated audio signals [14].

Historically, source separation has been approached using classical DSP techniques, such as independent component analysis (ICA) with blind source separation (BSS) [33, 34], non-negative matrix factorization (NMF) [35], and generalized wiener filtering (GWF) [36]. These methods laid the groundwork for the evolution of more advanced deep learning approaches explored in Section 2.1.1.

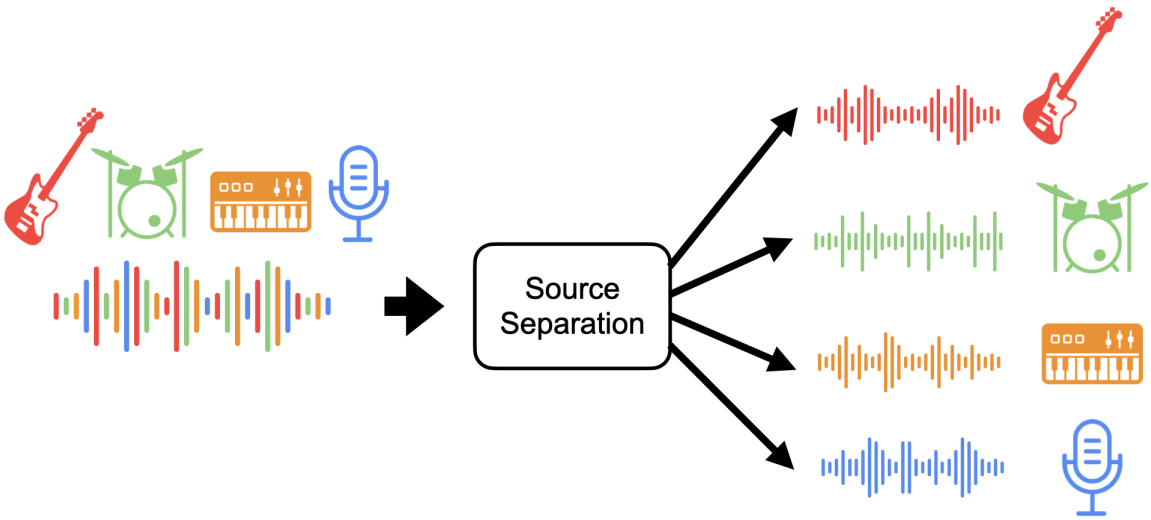



Figure 2.1: Illustration of extracting individual stems from a mixture.

Source: `https://source-separation.github.io/tutorial/landing.html`

### 2.1.1 Deep Learning Approaches for MSS

Deep learning, a subset of machine learning (ML), focuses on developing deep neural networks (DNN), which are computational models inspired by the human brain and consist of multiple hidden layers of interconnected nodes (or "neurons") that process data. During training, the model learns complex patterns by adjusting weights to minimize prediction errors, usually in the form of a unified loss function. A separate validation dataset is used in tandem and helps the training environment tune parameters to prevent overfitting. Finally, a smaller test dataset is used to evaluate the model's performance on new, unseen data, providing an unbiased measure of its effectiveness and generalization.

The rise of ML and deep learning in audio processing has advanced the field of MSS, overcoming challenges inherent in the classical DSP methods mentioned earlier. Deep learning models have the ability to learn complex temporal and spectral

representations of audio signals, enabling better separation performance, especially in challenging scenarios such as overlapping frequencies that may occur in the source stems [2, 9, 17, 18].

The following subsections provide a brief overview of various DNN model architectures that have defined the state-of-the-art (SOTA) in MSS, organized by the type of input audio domain they operate within and their unique contributions to the field. It is important to note that the architectures introduced in this literature review is not an exhaustive list of all researched and developed architectures, but rather serves to provide an overview of the state of the field.

#### 2.1.1.1 Spectrogram and Waveform Domain Models

To introduce domain-specific processing, time-frequency domain approaches encompass models that transform the audio into the frequency domain, typically using the short-time Fourier transform (STFT). Time-frequency approaches work on the same principles of image recognition models by leveraging the spectral representations of audio as spectrogram images, allowing models to better capture complex and overlapping spectral patterns over time frames. Models in this domain separate sources by applying predicted frequency masks to spectrogram time frames [17, 24].

While time-frequency approaches excel in capturing high-resolution spectral patterns, they come with inherent trade-offs such as artifacts introduced from time-frequency transformations, phase reconstruction errors, limited temporal precision, and high computational costs. Time-domain models operate directly on raw waveforms, bypassing the need for spectrogram transformations.

One of the earliest influential time-domain models is Wave-U-Net [26], which adapts the U-Net architecture to one-dimensional waveform processing (see Section 2.1.1.3). Wave-U-Net's use of downsampling and upsampling blocks captures features at different temporal scales directly from raw audio. By introducing linear interpolation for upsampling, it avoids aliasing artifacts and achieves competitive separation quality, laying foundational work for more complex models.

Demucs [18] further enhances the time-domain approach for music separation by adopting the U-Net encoder-decoder structure but introduces a bidirectional LSTM between the encoder and decoder that facilitates multi-scale feature learning for longer sequences of audio.

#### 2.1.1.2 Hybrid Time- and Frequency-Domain Models

Hybrid models have been introduced with the aim to combine the advantages of both time-domain and frequency-domain approaches. Improving from time-domain Demucs, Hybrid Demucs [19] keeps the same U-Net architecture but includes an additional processing path in the frequency-domain. By optimizing over the signal's raw waveform and spectrogram image, this approach has been shown to enhance separation performance for complex audio signals by leveraging the strengths of both domains in order to capture detailed temporal and spectral information.

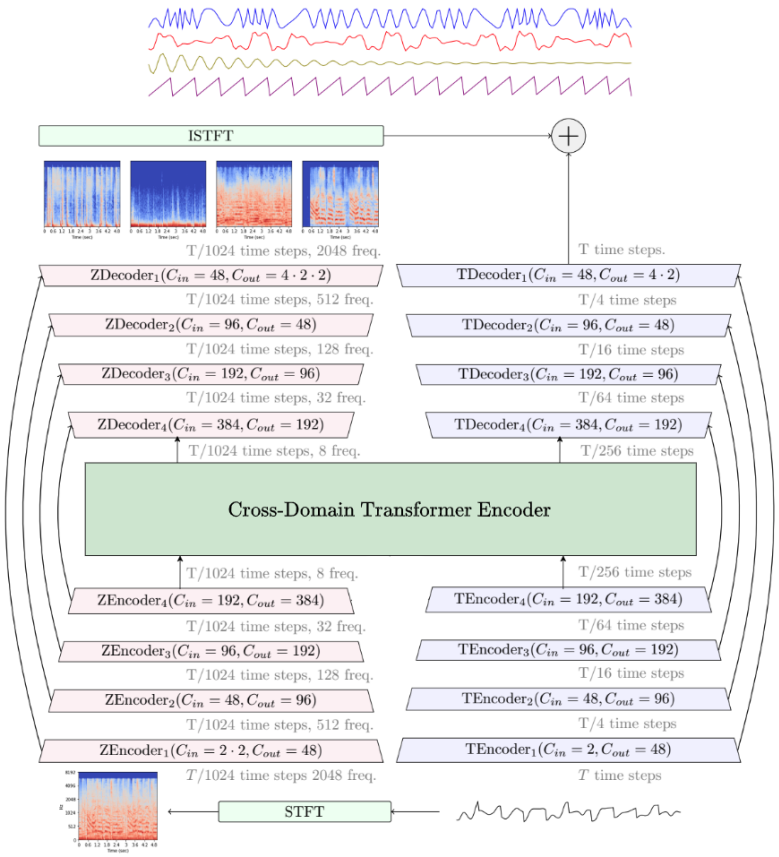



Figure 2.2: Diagram of the Hybrid Transformer Demucs [1].

#### 2.1.1.3 Autoencoders and U-Net Architectures

The autoencoder is a popular architecture that consists of convolutional encoder and decoder blocks that aim to reduce the input data into a lower-dimensional representation, to then reconstruct the data back into its original dimension from these learned representations. In audio applications, this involves compressing complex waveforms and spectrograms to a lower-dimensional feature space, where meaningful temporal and spectral patterns are extracted, and then reconstructing the audio to its original resolution while optimizing it for a specific task.

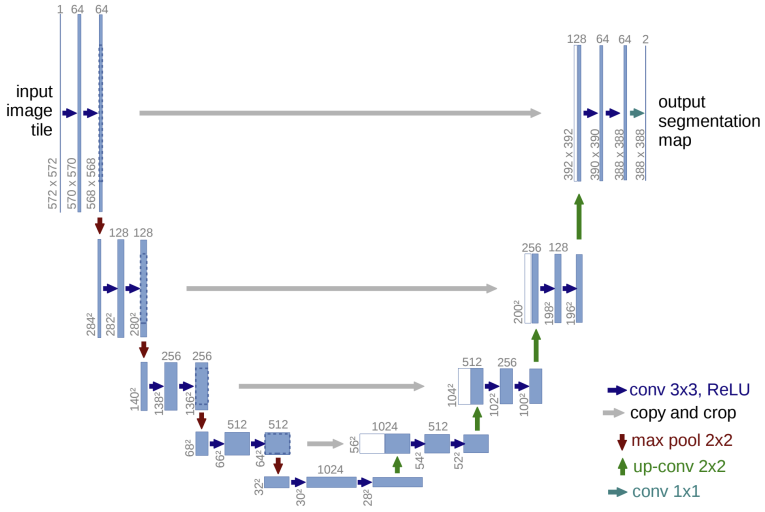



Figure 2.3: Diagram of the U-Net architecture, illustrating its encoder-decoder structure with skip connections [2].

A notable extension of this is the U-Net architecture, first introduced by [2] for biomedical image segmentation, which builds on the autoencoder structure by incorporating skip connections between the encoder and decoder blocks. These connections help preserve high-resolution features during reconstruction, improving the model's ability to retain fine details and spatial coherence.

The success of autoencoder-like architectures in image processing has also translated well to audio spectrograms, making them a popular choice for audio related tasks due to their ability to retain contextual and fine-grained details. Many MSS models expand on this foundational architecture, integrating enhancements such as hybrid domain processing [19], attention mechanisms [1, 4, 9], and additional latent space conditioning [37] to improve separation performance.

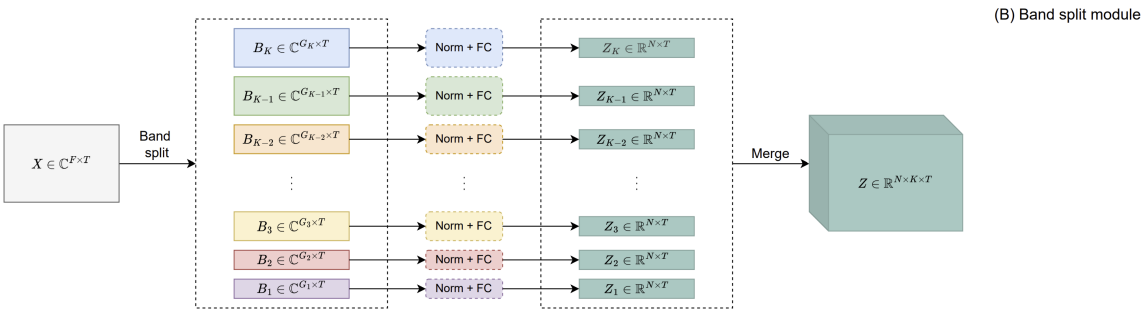



Figure 2.4: Diagram of the band-split module in [3].

#### 2.1.1.4 Recurrent Neural Network Models

While many of the models mentioned above rely on convolutional layers, U-Net architectures, or a combination of both, some architectures have turned to recurrent neural networks (RNNs) to better handle the challenge of processing longer sequences of audio data. Dual-path RNNs (DPRNN) have been proposed as an alternative to conventional 1-D CNN layers in TasNet to tackle the challenge of long sequence modeling by chunking input sequences and processing intra- and inter-chunk data separately [25]. This architecture enhances computational efficiency while preserving high-quality separation in the time domain.

#### 2.1.1.5 Band-Split and Multi-Band Spectrogram-Based Models

There is a recent emergence of frequency band-splitting techniques that outperforms the previous state-of-the-art models. The band-split RNN (BSRNN), introduces a novel frequency-domain method by splitting the spectrogram into subbands and using a stacked residual RNN to process both intra- and cross-band dependencies, achieving fine-grained frequency resolution and improving separation performance for complex music signals [3].

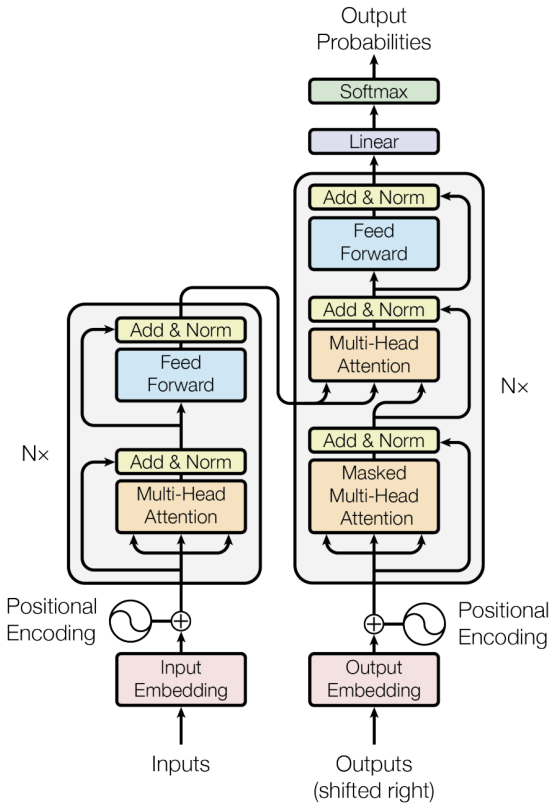



Figure 2.5: The Transformer model architecture from [4].

#### 2.1.1.6 Attention Mechanisms

Attention mechanisms in machine learning enable models to dynamically focus on the most relevant parts of input data, improving their ability to capture dependencies across long sequences. Originally introduced in natural language processing (NLP), attention plays a crucial role in handling sequential data, where input text is first tokenized into discrete units before processing. The Transformer is a sequence-based model that incorporates self-attention and positional encoding mechanisms to maintain long-range coherence in generation tasks in NLP [4] and image generation [38].

This concept extends to audio, where waveforms or spectrograms can be framed into smaller segments that act as tokens for the transformer-based attention layers to

maintain long-range context in music generation [39], and more recently, MSS tasks. Models like Hybrid Transformer Demucs (HTDemucs) and Latent Source Attentive Frequency Transformation (LaSAFT) integrate transformer layers into their architectural structures, replacing the biLSTM traditionally used for sequential data processing [1, 17]. By incorporating attention mechanisms, transformer-based attention mechanisms significantly outperform other architectures in capturing and maintaining long-term contextual information.

The Band-Split Rotary Position Embedding Transformer (BS-RoFormer), further demonstrates the efficacy of Transformers in sequential data modeling by combining them with band-split modules for more accurate and efficient handling of spectral data in MSS [9].

### 2.1.2 Open Source Datasets for MSS

The development and success of MSS models heavily rely on large-scale datasets like MUSDB18 [40], widely considered the benchmark for evaluating model performance. MUSDB18 contains 150 full-length tracks, totaling approximately 10 hours of audio across genres, with ground-truth stems for four categories: vocals, drums, bass, and other. The dataset is divided into 100 training tracks and 50 test tracks, all in stereo and encoded at a 44.1kHz sampling rate. An uncompressed alternative, MUSDB18-HQ [41], is also available for models targeting wider bandwidth audio, offering WAV files that extend to 22kHz content while MUSDB18 only extends to 16kHz. Due to these ground-truth classes, most models trained on MUSDB18 produce the same four separated outputs.

While other datasets, such as MoisesDB [42], expand on these options by including 240 tracks across 45 artists and twelve musical genres with more granular source categories (e.g., five and six stems), the limited availability of large, open-source datasets remains a significant challenge in the field of MSS.

The scarcity of open-source datasets remains a major pain point in the MSS field of hindered standardization and slow academic progress. Much of the innovation in source separation is currently driven by private companies with access to proprietary multitrack data, leading to models that are often not reproducible. For example, HTDemucs was trained on an additional 800 songs beyond the MUSDB18 dataset [1], and Spleeter leveraged over 20,000 tracks for training [43]. Furthermore, the legality of data ownership in many of these cases remains unclear, raising ethical and legal concerns about the use and distribution of these pre-trained models.

## 2.2 Automatic Stereo-to-Multichannel Upmixing

### 2.2.1 Overview of Multichannel Surround Sound

Multichannel audio configurations, such as 5.1 and 7.1, are designed to create enveloping audio experiences by extending beyond regular stereo audio. These formats are commonly used in home theaters, cinemas, and gaming environments to recreate spatial soundscapes by distributing audio across multiple channels, including left, right, center, low-frequency effects (LFE), and surround channels [12]. The accurate reproduction of these channels is essential for providing listeners with a realistic and immersive auditory experience.

### 2.2.2 Definition of Upmixing

In the context of audio and music production, *upmixing* refers to the process of converting an audio signal from its original format to a larger configuration with more channels [44]. Conversely, converting an audio signal from a larger format to a smaller one with fewer channels is known as *downmixing*. Upmixing can be categorized into "informed" upmixing, which uses prior information or metadata to guide the construction of the upmixed output [45], and "blind" upmixing which aims to create the output purely from the input signal without the use of side information [44].

### 2.2.3 Available Solutions on the Consumer Market

There already exists surround upmixing solutions available on the market that are capable of converting stereo content to surround sound. These solutions include consumer electronics such as audio/video receivers (AVRs) integrated with Dolby Pro Logic II [11] designed for home theater setups, and software plugins [46] for use in digital audio workstations (DAWs).

### 2.2.4 Classical DSP Approaches to Stereo Upmixing

Proposed methods for upmixing stereo audio to multichannel formats have included techniques such as the "passive" and "adaptive" surround decoding matrix [11], primary-ambient extraction (PAE) [10,16,47], mid-side (M-S) decomposition [48], and center channel derivation [6].

| **Matrix** | **Left** | **Right** | **Center** | **Rear Left** | **Rear Right** |
|---|---|---|---|---|---|
| Left | 1 | 0 | $\sqrt{\frac{1}{2}}$ | $j\frac{\sqrt{3}}{2}$ | $j\frac{1}{2}$ |
| Right | 0 | 1 | $\sqrt{\frac{1}{2}}$ | $k\frac{1}{2}$ | $k\frac{\sqrt{3}}{2}$ |

$j = +90°$ phase shift, $k = -90°$ phase shift.

Table 2.1: Dolby Pro Logic II Decoding Matrix from [11].

#### 2.2.4.1 Dolby Surround Decoding

Dolby Surround Decoding is based on a predefined processing matrix, where additional surround channels are derived from a standard stereo signal through a combination of panning, phase manipulation, difference signals, and filtering techniques. There are two primary decoding methods: passive and adaptive. Passive decoding simply applies a fixed matrix to extract surround information as it exists in the stereo mix, without altering the levels dynamically. Adaptive decoding, on the other hand, employs gain riding and signal cancellation mechanisms to resolve a dominance vector, which determines the most prominent directional component of the sound field for playback. This helps improve separation between channels and enhances spatial accuracy. Dolby Pro Logic II (DPLII) extends the original Dolby Surround Decoding method by using a refined matrixing approach that improves channel separation and stereo width in the surround field. As shown in Table 2.1, the DPLII matrix applies phase shifts of +90° (j) and -90° (k) to generate the surround channel information [11].

#### 2.2.4.2 Primary-Ambient Extraction

Primary-Ambient Extraction (PAE) is a popular technique used in upmixing that operates on the assumption that a stereo signal model is comprised of primary and

ambient components. The primary components of a signal consist of direct and correlated sounds that are prominently shared between stereo channels. This includes primary instrument stems and prominent audio signals such as vocals, drums, and bass. In contrast, the ambient component is comprised of diffuse and decorrelated signals, such as reverb and delay, which are less common between the two stereo channels.

PAE has emerged as a preferred approach for upmixing because it effectively allows for the expansion of the perceived spatial field of stereo audio by routing the ambient signal to the additional surround channels of a multichannel surround system. In a 5.1 surround speaker configuration, this would be the rear left and right speakers. Various techniques have been developed to achieve this decomposition of direct and diffuse signals, including methods based on frequency-domain inter-channel coherence [13], cross- and auto-correlation [14], principal component analysis (PCA) [15,16], frequency-domain normalized least-mean-square (NLMS) adaptive filtering, and time-domain least-mean-square (LMS) filtering [8, 49].

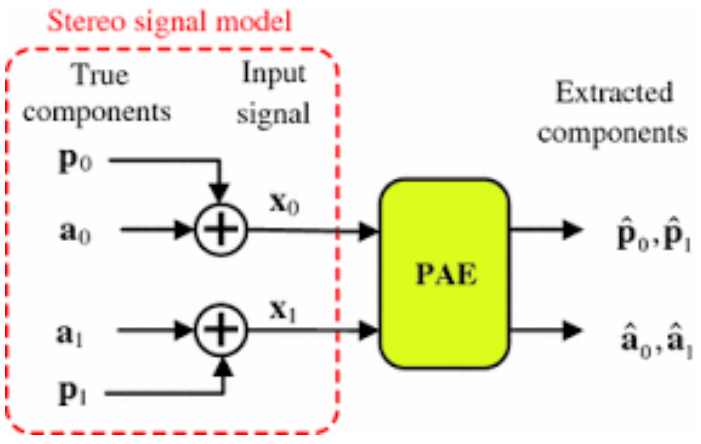



Figure 2.6: Stereo signal model for PAE signal components [5].

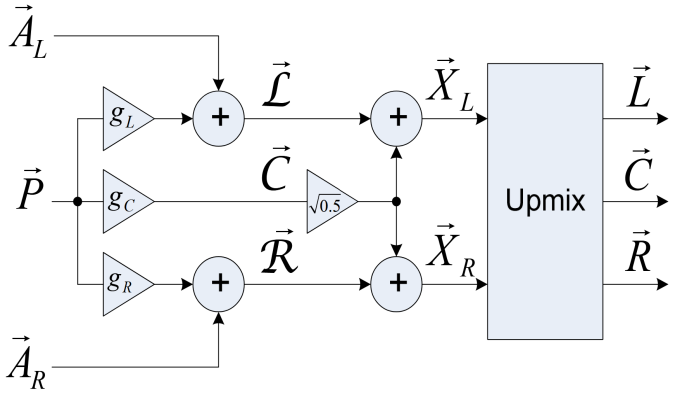



Figure 2.7: Three-channel signal model showing the mixture of primary ($\vec{P}$) and ambient ($\vec{A}$) sources that form the $\vec{L}$, $\vec{C}$, $\vec{R}$ channels [6].

#### 2.2.4.3 Center Channel Derivation

Center Channel Derivation or Decomposition (CCD) leverages the psychoacoustic phenomenon of the "phantom center," where sounds located in the center of a stereo image are perceived as originating in-between the physical stereo speakers [6]. CCD extends the PAE stereo signal model and works off of the assumption that the primary-ambient signal components are distributed amongst the three front channels, where the center channel contains only the primary signal, and the left and right channels contain a mixture of the corresponding channels primary and ambient components (see Figure 2.7). With an adapted three-channel primary-ambient signal model, the CCD algorithm applies vector-based signal decomposition in the frequency domain to geometrically estimate a dedicated center channel. This dedicated center channel mitigates acoustic cross-talk issues inherent in the phantom center effect, where identical sounds from both speakers create phase cancellations that can introduce comb-filtering in stereo audio playback. By separating the center into a physical channel, CCD reduces cross-talk and improves audio clarity and

dialogue intelligibility in multichannel audio applications [50]. This algorithm is not only effective for generating the front channels in a multichannel surround format, but may also enhance ambient extraction quality for the surround channels in a 5.1 upmix by serving as a preprocessing step to remove correlated primary components prior to ambient signal extraction (see Section 3.2.1).

### 2.2.5 Deep Learning Approaches to Upmixing

The use of DNNs for upmixing has been explored in prior work that focuses primarily on achieving PAE through neural networks and applies additional post-processing to assist in panning and distribution of the signal appropriately [10, 30, 31, 44].

One method uses a supervised learning approach to extract ambient components from mono recordings, modeling primary-ambient separation as a classification problem over frequency bins [44]. A similar DNN-based approach classifies frequency-frame bins as either direct or ambient sound using a fully-connected multi-layer perceptron (MLP) [10]. Another method trains a network on two-to-five channel surround decoding, optimizing spectral weights to approximate spatial properties of surround sound for enhanced performance over traditional methods [30]. Another similar approach implements subband-based upmixing to learn the features of the center and rear surround channels from the multichannel input [31].

While these methods focus on blind upmixing through PAE, a unique approach leverages a variational autoencoder (VAE) to learn the spatial distribution of musical instruments in a source 5.1 mix, mapping these locations in a latent space [37]. This model then upmixes the target input stereo audio through style transfer, applying

the learned spatial locations of instruments onto the stereo input through directional panning. Although not directly performing source separation task, this paper still suggests the possibility for ML architectures to possess the ability to capture and learn complex spatial features of audio in the context of upmixing.

### 2.2.6 Voice-Primary-Ambient Extraction Upmixing

With recent advancements in MSS, emerging research has begun to integrate MSS architectures (discussed in Section 2.1.1) into upmixing frameworks. In the context of stereo-to-5.1 upmixing, Voice-Primary-Ambient Extraction (VPA) assigns the output of a pretrained MSS model as the primary component for the front channels, while some form of ambient signal extraction is used to generate the surround channels. The VPA method was first proposed by Samsung Research, using the OpenUnmix vocal extraction model [27] to isolate vocals in the center channel, assign residual stems to the left and right channels, and apply the Equal-Levels Ambience Extraction (ELAE) algorithm [14] to generate ambient signals for the rear channels (see Figure 2.8).

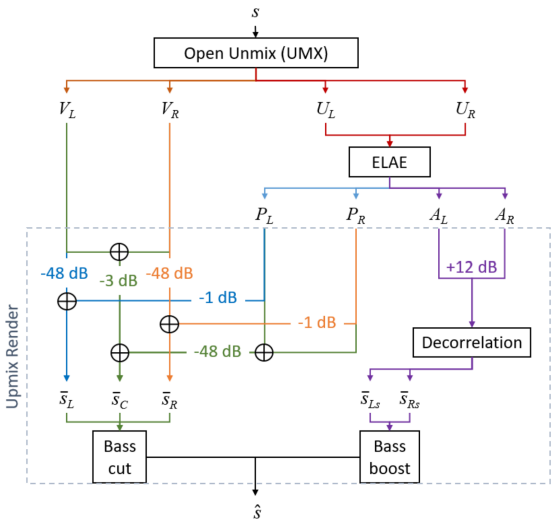



Figure 2.8: VPA upmixing schema as presented in [7].

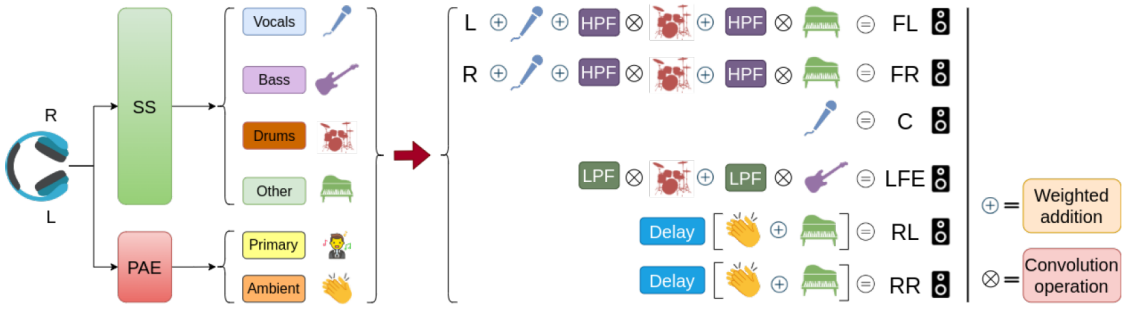



Figure 2.9: VPA upmixing schema as presented in [8].

Similarly, VanceAI [8] builds upon the VPA method by replacing the vocal extraction model with a more advanced SOTA architecture, such as Demucs [18], to separate and remix the primary sources. For ambience extraction, it employs a Normalized Least Mean Squares (NLMS) adaptive filter in combination with an Adaptive Panning (ADP) algorithm (see Figure 2.9). Most notably, this adaptive filtering technique optimizes a set of weights from the stereo input, which are subsequently refined using the ADP algorithm. However, the requirement to optimize and fine-tune the adaptive filter weights makes this method less suitable for real-time applications, favoring offline processing instead.

## 2.3 Summary of Literature Review

Section 2.1 introduces the field of MSS, defining it as the process of decomposing a mixed audio signal into its individual components. Various applications are outlined, from vocal extraction to upmixing, and the section highlights the challenges associated with separating highly correlated and overlapping audio sources.

Section 2.1.1 discusses deep learning approaches for MSS, providing an overview of key architectures categorized by input domain. Major architectural advancements that address processing and contextual modeling highlight the shift toward more sophisticated neural networks capable of overcoming the limitations of classical DSP methods.

Section 2.2 reviews stereo-to-multichannel upmixing approaches, covering both commercial solutions and technical methods. Subsections 2.2.4.2 & 2.2.4.3 emphasize PAE and CCD as foundational approaches for decomposing stereo signals into the necessary direct and ambient components that form the basis for many blind upmixing strategies.

Subsection 2.2.5 presents deep learning approaches specifically optimized for PAE and other spatial decomposition tasks. These methods achieve notable improvements in automatic upmixing with reduced reliance on DSP, and include innovations such as binary classification of ambient signal frames and spatial modeling via variational autoencoders for style transfer.

Finally, Subsection 2.2.6 explores VPA upmixing methods, which leverage pre-trained MSS models to separate primary stem components alongside DSP-based algorithms for ambient signal extraction. These methods represent a recent trend toward more efficient, modular, and perceptually coherent upmixing pipelines that directly leverage high-performance source separation models.

# CHAPTER 3

# Methodology

This chapter outlines the aims and objectives of the thesis, defining the scope of the problem and positioning the proposed solution within the current state-of-the-field as reviewed in Chapter 2. It includes an overview of the identified knowledge gap, research motivation, and objective, followed by a detailed experimental setup for implementing the proposed upmixing solution. This setup covers dataset procurement, data preparation, training process, and hardware configurations. For reproducibility, refer to Appendix A for links to the code repository, model weights, datasets, and audio examples.

## 3.1 Knowledge Gap and Research Motivation

Many approaches discussed in Section 2.2.5 rely on simpler DNN architectures that lack the advanced feature extraction and separation capabilities of the more intricate MSS models presented in Section 2.1.1. Although the improved methods in Section 2.2.6 incorporate more recent MSS models into the upmixing system, they still rely heavily on DSP-based methods for generating the surround channels.

As informed by the literature review, a clear knowledge gap exists in the development of an automatic blind upmixing system that leverages advanced DNN

architectures to generate both direct and ambient signal components for the front and surround channels, respectively. Addressing this gap by investigating the task of PAE with advanced DNNs expands the field of audio source separation by introducing ambient signal extraction as a specialized separation task within the context of music and entertainment audio. In its specific application to stereo-to-surround upmixing, it also has the potential to significantly improve upon current approaches by integrating deep learning into all components of the upmixing system.

This leads to this paper's central research question: How can advanced MSS architectures be adapted for ambient signal extraction to enable an enhanced blind stereo-to-surround upmixing system?

### 3.1.1 Proposed Solution and Hypothesis

Among the various approaches to automatic blind upmixing, the method best aligned with our proposed system follows the PAE-based VPA framework (explored in Section 2.2.6) to derive the signal components needed for the final upmix. To evaluate the effectiveness of SOTA MSS models for this task, selected architectures introduced in Section 2.1.1 will be trained on a custom dataset of primary and ambient music audio (see Sections 3.2.2–3.2.5) and compared against reference methods [7,10] using a controlled subjective listening study (Section 4.2) and objective signal reconstruction metrics (Section 4.3).

We hypothesize that utilizing higher-complexity MSS models will improve performance in capturing nuanced audio features for the PAE task, compared to the approaches discussed in Section 2.2.5, and will offer greater generalizability across diverse audio content than the DSP-based methods presented in Section 2.2.4.2.

## 3.2 Experimental Setup

### 3.2.1 The Upmixing System Design

Figure 3.1 details the processing pipeline of the proposed upmixing system, from the stereo input to the final concatenation of the multichannel output. Notably, our proposed system further expands upon the VPA upmixing approach by combining MSS and CCD for generating the front three channels, along with an MSS model specifically optimized to extract the ambient components used for the surround channels in the final upmix.

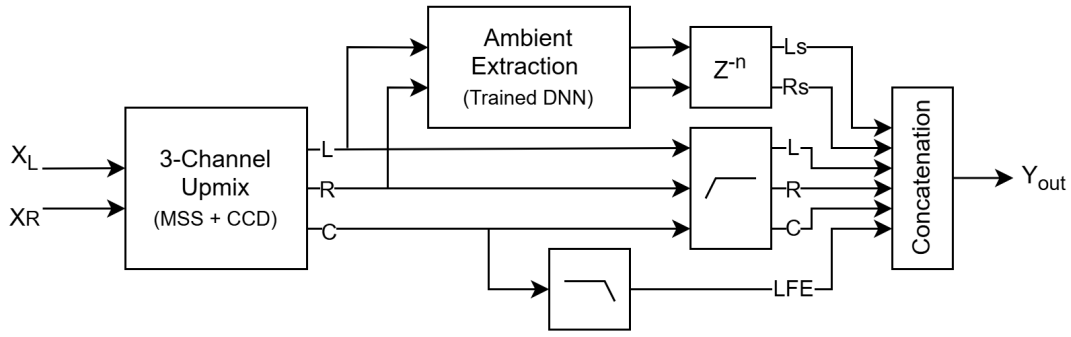



Figure 3.1: Block diagram of the proposed upmixing schema.

I. **Three-Channel Upmix Block**

The Three-Channel Upmix block is responsible for processing the raw stereo input signal to separate the left, center, and right components of the stereo signal into the front channels. This block employs a combination of MSS and the CCD algorithm [6] to perform a two-to-three channel upmix of the primary signals components, following a strategy similar to the VPA methods discussed in Section 2.2.6.

II. **Ambient Extraction Block**

The Ambient Extraction block leverages an MSS model optimized to separate ambient components from the processed left and right signal outputs of the Three-Channel Upmix block. An additional 12-millisecond delay is applied to the extracted ambient signal via a zero-padding operation to enhance perceived spatial separation between the front and rear channels and to mitigate potential phase interference with signals from the front three channels.

III. **Additional Filtering Blocks**

A final step applies low-pass and high-pass filtering to generate the low-frequency effects (LFE) channel and to remove low-frequency content overlap from the five surround channels. This is achieved using a low-pass biquad filter with a cutoff frequency of 200 Hz for the LFE channel, and a high-pass biquad filter with a cutoff frequency of 150 Hz applied to the remaining channels.

### 3.2.2 Dataset Procurement and Data Preparation

For our training dataset, we utilize the open-source MUSDB18-HQ [41] and MoisesDB [42] MSS datasets, providing a combined total of 390 individual songs sampled at 44.1 kHz. The combined dataset undergoes verification through an automated script, which checks for improper labeling, silent tracks, inconsistent track lengths, and other discrepancies in the audio data, discarding any ambiguous or invalid entries. Following this initial data procurement and cleaning process, a total of 385 valid song tracks remain, accumulating to 1,925 individual stereo wave files and approximately 120 hours of audio data

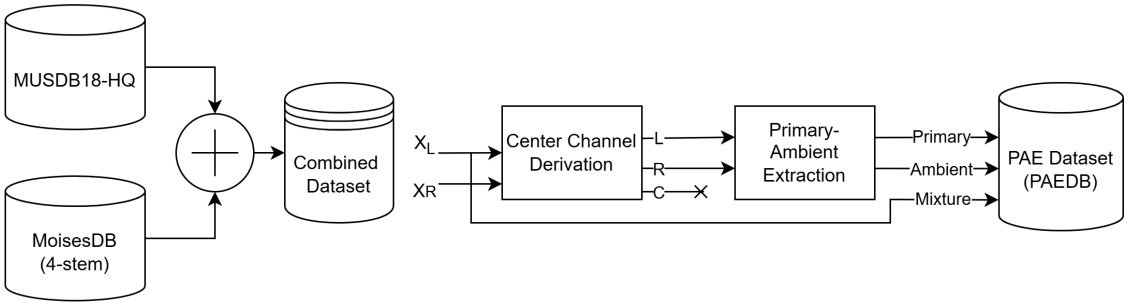



Figure 3.2: Diagram illustrating the preparation and generation process of the custom Primary-Ambient Extraction Dataset for Music (PAEDB).

As shown in Figure 3.2, we construct a custom music dataset specifically for training tasks involving PAE-based source separation, referred to as PAEDB. Each entry in PAEDB consists of the original mixture file along with corresponding primary and ambient stems, extracted using the frequency-domain correlation-based ELAE method [14], with additional preprocessing of the left and right components using the CCD algorithm [6] to enhance ambient extraction quality (see Section 2.2.4.3 for details). To expand the total dataset size, each individual stem from the base 4-stem dataset (vocals, drums, bass, other, and mixture) is treated as a unique entry. After generating the corresponding primary and ambient components for each, the target stem is repurposed as the new mixture file. This process effectively increases the trainable dataset size by approximately `x5`. After generating the PAE components and performing additional validation of the audio data, the final PAEDB consists of 1,809 unique tracks, resulting in 5,427 individual stereo stem files and approximately 550 hours of audio, totaling around 150 gigabytes.

### 3.2.3 Model Architecture Selection

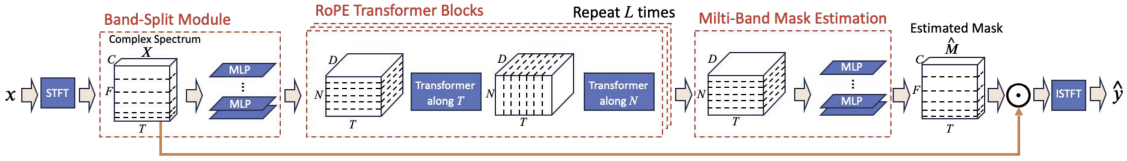



Figure 3.3: The framework of the Band-Split RoFormer system as presented in [9].

To evaluate the effectiveness of high-complexity MSS architectures for PAE in upmixing, we implemented two distinct models trained on the PAEDB dataset. The first selected model is the Mel-RoFormer, developed by ByteDance Research [9]. This model employs a Mel-spectrogram transform in a band-split RNN and utilizes hierarchical Transformer with Rotary Position Embeddings to model both inner- and inter-band sequences for multi-band mask estimation. As of this thesis, the Mel-RoFormer achieves SOTA SDR performance on the MUSDB18-HQ benchmark [41, 51], and was thus chosen as the high-complexity MSS model to be optimized for PAE.

The second model is a simpler multi-layer perceptron (MLP) architecture originally proposed in [10]. This model consists of three fully connected linear layers and operates by flattening the real and imaginary components of the input STFT representation into a one-dimensional feature vector. It performs binary classification of primary and ambient signal frames, applying the softmax predicted probability (between 0 and 1) as a gain factor to effectively act as a dynamic gate that attenuates primary signals while preserving ambient components. This model serves as a reference comparison to determine whether a simpler approach is effective at ambient extraction. Since this architecture was explicitly designed for PAE, this approach provides a direct reference and evaluation of the deep learning method proposed in [10].

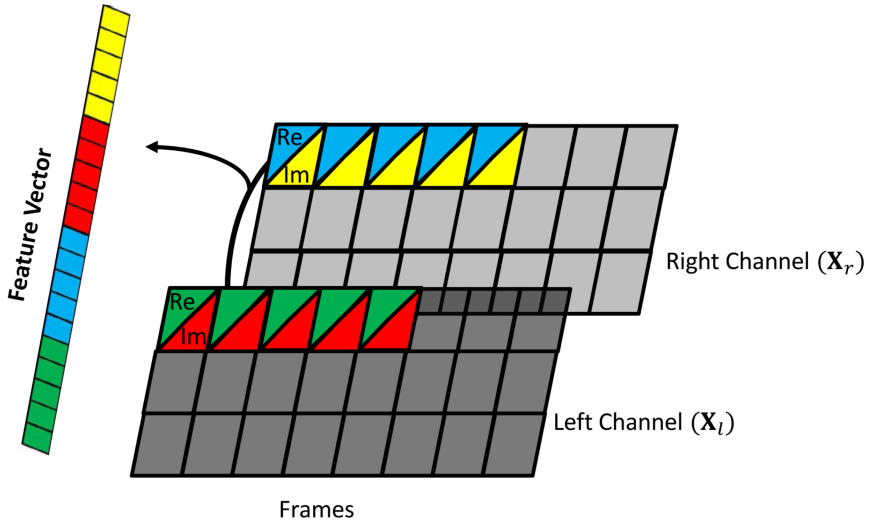



Figure 3.4: Preprocessing logic for flattening input STFT feature vectors for the MLP, as presented in [10].

### 3.2.4 Model Training Parameters

Our training procedure follows a supervised learning training strategy over the prepared PAEDB audio dataset. For simplicity and initial experimentation, the training parameters are set as follows, using the same configurations reported in the respective literature for each model. See Section 5.1 for the results of the training runs.

1. **Mel-RoFormer** – The optimizer used is AdamW, with a learning rate of 5e-5 and an EMA momentum of 0.999 for stabilizing parameter updates. Training employs a batch size of 3 (effectively doubled to 6 with the dual-GPU setup), with gradient accumulation steps set to 1 and mixed precision enabled for improved computational efficiency. The training run is set for 100 epochs for 4,000 steps, with a learning rate reduction applied after 2 epochs of no improvement in the monitored metric, using a reduction factor of 0.95 to fine-tune the optimization and prevent overfitting.

2. **3-Layer MLP** – The model is trained using the AdamW optimizer with a learning rate of 1e-3, batch size of 16, and 200 training epochs. Mixed precision training is enabled, and learning rate reduction is applied with a patience of 5 epochs and a reduction factor of 0.95. The network comprises three linear layers with ReLU activations, reducing the feature dimensionality through hidden layers of sizes 15 and 10 before outputting a single probability through a sigmoid activation, representing the presence of ambient content.

The loss function for training the Mel-RoFormer model combines L1 loss for waveform reconstruction in the time domain with a weighted multi-resolution STFT (MRSTFT) loss to maintain fidelity across multiple time-frequency resolutions, following the approach in [9]. In contrast, the MLP model uses only L1 loss applied to the final softmax layer output, with binary labels assigned as 0 for primary stems and 1 for ambient stems, consistent with the implementation in [10].

Data augmentation is applied through random loudness scaling between 0.1 and 1.0 to improve generalization. Additionally, multi-threaded data loading is enabled using eight parallel processes to optimize training performance.

### 3.2.5 Hardware and Software Training Resources

The training process is conducted on an Ubuntu 22.04.4 LTS (Jammy Jellyfish) operating system using a dual-GPU setup with two NVIDIA A10 GPUs, each equipped with 23 GB of VRAM, running on CUDA version 12.5.

The CPUs are dual Intel Xeon Silver 4316 processors, offering a total of 80 cores (40 physical cores, 2 threads per core) with a maximum clock speed of 3.4 GHz. The

system includes 30 GB of RAM, with approximately 25 GB available during training, and an additional 8 GB of swap memory.

Training scripts and model definitions utilize PyTorch Version 2.5.0 with mixed precision enabled to improve computational efficiency. The dual-GPU setup effectively doubles the batch size by distributing workloads across two instances of the model mapped to each GPU which enables faster data loading and the ability to handle larger datasets.

# CHAPTER 4

# Evaluation and Performance Metrics

This chapter presents the evaluation methodology used to assess the perceptual and objective performance of the proposed upmixing systems. It includes a description of the evaluation dataset, the design and procedure of a controlled user listening study, and the quantitative analysis using standard signal reconstruction metrics. Both subjective and objective evaluations are used to compare the trained models against existing DSP- and DNN-based upmixing methods. See Sections 5.2 and 5.3 for the corresponding results, and Appendix A for the evaluation scripts.

## 4.1 Evaluation Test Set

The evaluation set consists of 12 full-length stereo music tracks sourced from YouTube, each ranging from 2 to 6 minutes in duration and sampled at 44.1 kHz. The audio was acquired using a Python-based tool developed for this project [52], which wraps the `pytube` Python module [53] to simplify downloading audio content without needing to use the official YouTube API. The tool extracts the highest-quality available audio stream (128 kb/s, `.m4a` format), which is then converted to 44.1 kHz `.wav` format using `ffmpeg` [54].

For the subjective evaluation, a subset of six tracks was manually curated from the full dataset. These samples were chosen to reflect a diverse range of genres, production styles, energy levels, and ambient textures, with particular attention to variation in stereo mixing practices.

The complete 12-track dataset was used for the objective evaluation to enable a more comprehensive and statistically robust comparison. All audio content was used strictly for academic research purposes and is not intended for redistribution or commercial use.

## 4.2 Subjective Evaluation via User Listening Study

To assess the perceptual quality of the trained models in the context of audio upmixing, a user listening study was conducted. The study aimed to evaluate spatial characteristics, immersion, and spectral fidelity of the processed audio excerpts within a controlled experimental setup.

### 4.2.1 Participant Demographics and Screening

For the user study, 12 participants were recruited to evaluate audio excerpts processed by four different upmixing systems (see Section 4.2.3). Prior to the listening session, participants completed a brief demographic questionnaire, reporting their age, gender, experience with audio equipment, and self-assessed hearing ability. To ensure reliable responses, all participants were required to confirm normal hearing or disclose any known impairments.

Additionally, a cognitive immersion assessment was included at the end of the study. It consisted of several questions aimed at gauging participants' general ten-

dency to become absorbed in immersive experiences, including their self-reported awareness and sensitivity to spatial cues in their environment.

### 4.2.2 Ethical Considerations and Participant Consent

The study and its adherence to ethical guidelines were approved by the Institutional Review Board (IRB) at the University of Miami. Participants were informed about the purpose, procedures, potential risks, and confidentiality measures of the study through an informed consent form. They were explicitly notified that participation was voluntary and that they could withdraw at any time without consequence. To minimize listening fatigue, participants were also permitted to take short breaks between trials and adjust playback volume for comfort.

### 4.2.3 Stimuli and Systems

Six distinct 1-minute excerpts were selected from a larger test set to capture a diverse range of audio content. These samples were curated to reflect factors that would potentially influence the upmixing process such as genre, energy, ambience, and stereo mixing style. The evaluated stimuli are listed below:

- **Stimuli A:** Sound & Color – Alabama Shakes (Blues/Soul Rock) – A sparse mix with swirling ambient metallophone, reverberant vocals, and dynamic instrumentation [55].

- **Stimuli B:** Let You Break My Heart Again – Laufey & Philharmonia Orchestra (Orchestral Jazz) – A jazz ballad with orchestral arrangement, acoustic guitar, and vocals [56].

- **Stimuli C:** Snow Song – Adrianne Lenker (Acoustic Folk) – Acoustic guitar with natural room ambiance and spatially panned background vocals [57].

- **Stimuli D:** Take Five – The Dave Brubeck Quartet (Instrumental Jazz) – A classic jazz piece featuring saxophone, piano, bass, and drums, presented in a tight spatial mix [58].

- **Stimuli E:** Money – Pink Floyd (Progressive Rock) – Busy and energetic electric guitar solos with highly distorted and reverberant sound effects [59].

- **Stimuli F:** Prison Break Scene – Guardians of the Galaxy (Cinematic Audio) – A film scene with dialogue, sound effects, and an orchestral score [60].

Each stimulus sample was processed through all four upmixing systems and played back to participants using a calibrated 5.1 surround sound system. To mitigate order effects and bias, the six stimuli were first randomized to avoid any fixed playback sequence. Within each stimulus, the presentation order of the four systems was further shuffled to prevent order bias toward a particular upmixing system. The evaluated systems are listed below:

- **System A:** All-channel Stereo (Anchor) — The original stereo mix duplicated and played back through the left (L), right (R), back left (Ls), and back right (Rs) speaker channels. This serves as a baseline system, similar to those used in [8, 37].

- **System B:** DSP-Based (Reference) — Implements the VPA upmixing method from [7], using DSP-based PAE [14] to derive the additional surround channels.

- **System C:** DNN-Based (Our System) — A VPA upmixing method using the optimized Mel-RoFormer model [9], trained on the PAEDB dataset to extract ambient signals for the surround channels.

- **System D:** MLP-Based (Reference) — A VPA upmixing method using the 3-layer MLP model from [10], also trained on the PAEDB dataset.

### 4.2.4 Study Procedure

This study followed a MUSHRA-style test design, in which participants rated each upmixing condition relative to the others for each stimulus. The listening sessions were conducted in a controlled environment equipped with a calibrated 5.1 surround sound speaker array to ensure consistent spatial audio reproduction. Each participant listened to a total of 24 stimuli, comprising six distinct audio excerpts processed by each of the four upmixing systems. The full session was designed to last approximately 45 minutes.

### 4.2.5 Data Collection

Each audio excerpt was evaluated using a 7-point labeled Likert scale across three perceptual dimensions:

1. **Overall Envelopment and Immersion (Preference)** — Assesses how fully the listener perceives being surrounded by and engaged in the sound field.

2. **Spatial & Temporal Quality (Realism of Acoustic Space)** — Evaluates the localization, depth, and timing accuracy of sound sources within the spatial mix.

3. **Spectral Quality (Coherence & Presence of Artifacts)** — Measures timbral fidelity, the presence of distortions, and overall spectral balance.

## 4.3 Objective Metric Evaluation

In addition to the subjective listening study, objective metrics were used to assess the separation and reconstruction quality of the upmixed audio. These evaluations are essential for quantifying signal quality and comparing the performance of different upmixing systems.

For this analysis, we focus on comparing System C (Mel-RoFormer) and System D (MLP), using the DSP-generated signals as reference ground truths. Since the PAEDB dataset was constructed using ELAE [14] (as described in Section 3.2), these reference signals serve as the baseline for evaluating how accurately each neural network reproduces the target ambient signal. Thus, the following objective metrics are used to evaluate system performance:

- **Signal-to-Distortion Ratio (SDR)**: Measures the overall quality of the separated signals by computing the ratio of target signal energy to distortion, including noise, interference, and artifacts. Higher SDR values indicate better separation quality.

- **Scale-Invariant Signal-to-Distortion Ratio (SI-SDR)**: A variant of SDR that eliminates dependencies on absolute signal amplitude, providing a more robust evaluation of separation quality by focusing on distortions relative to the target signal.

- **Log-Spectral Distance (LSD)**: Assesses spectral fidelity by computing the logarithmic distance between the estimated and ground-truth power spectra. Lower LSD values indicate more accurate spectral reconstruction.

### 4.3.1 Metrics Processing Procedure

To automate the objective evaluation process, a Python script was developed to iterate over each corresponding pair of reference and estimated audio files. For each excerpt, metrics were computed using 45-second analysis windows with a 5-second hop size, producing a sequence of frame-wise evaluations across the duration of each track.

For each excerpt, metric scores were averaged across all analysis windows to yield a per-track result. An overall mean was then computed across the full 12-track evaluation set to produce a system-level performance metric.

# CHAPTER 5

# Results

This chapter presents the results of the training process, subjective listening study, and objective evaluation metrics for the proposed ambient signal extraction methods in the upmixing systems. The analyses include both statistical and perceptual comparisons of Stimuli A–F, each upmixed using Systems A–D. All systems and stimuli are referred to according to the labeling scheme defined in Section 4.2.3.

## 5.1 Model Training Results

To evaluate the training progress of the Mel-RoFormer and MLP models integrated into Systems C and D, respectively, we track the training loss and resource utilization across 100 epochs. All metrics, training graphs, and experiments were logged using the Weights & Biases experiment tracking API [61].

### 5.1.1 Training Loss Curves

Figure 5.1 shows the training loss convergence for both models over the course of 100 training epochs. System C demonstrates a steady decline in loss, indicating stable convergence despite the use of a combined L1 and MRSTFT loss, which typically

results in higher absolute loss values. In contrast, System D stabilizes much earlier, suggesting limited capacity to learn complex ambient extraction patterns. Its training curve is less stable, exhibiting multiple plateaus at local minima before continuing to descend toward a global minimum – or at least the lowest point observed within the 100 training epochs.

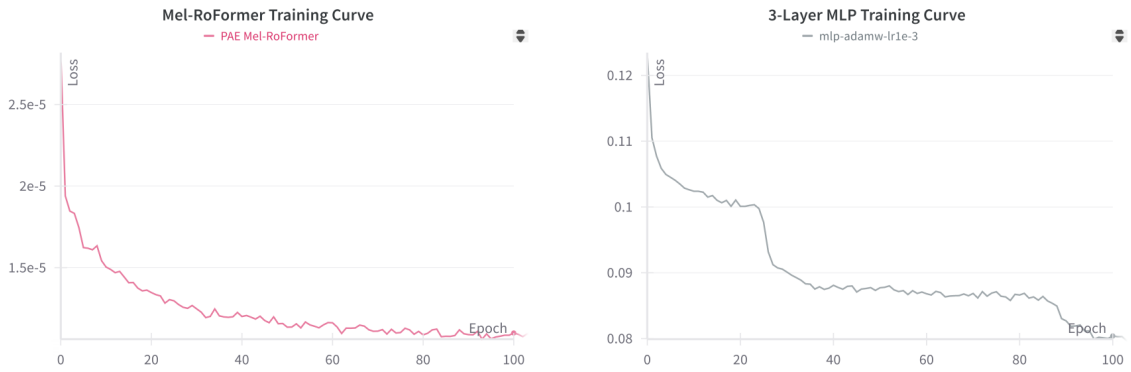



Figure 5.1: Training loss curve for Systems C and D.

### 5.1.2 Resource Utilization Comparison

To evaluate the computational efficiency of both models, we also compare their GPU memory allocation over time during training. Figure 5.2 presents the percentage of allocated GPU memory throughout the training process for Systems C and D.

System D consistently utilized approximately 27% of the GPU memory, whereas System C exhibited a utilization range of approximately 17-24%. However, it is important to note that System C was trained using a lightweight implementation of the Mel-RoFormer architecture and employed data parallelism across two GPUs, effectively doubling the actual resource usage depicted in the graph.

Another significant difference lies in the training duration required to complete 100 epochs. System D completed training in approximately 42.4 hours ($\approx$ 1.77 days),

whereas System C required approximately 249.5 hours (≈ 10.4 days), nearly six times longer. This highlights the trade-offs between model complexity and computational efficiency. While the Mel-RoFormer model achieves superior separation quality, its significantly longer training time and greater GPU resource demands must be considered when scaling for real-world applications.

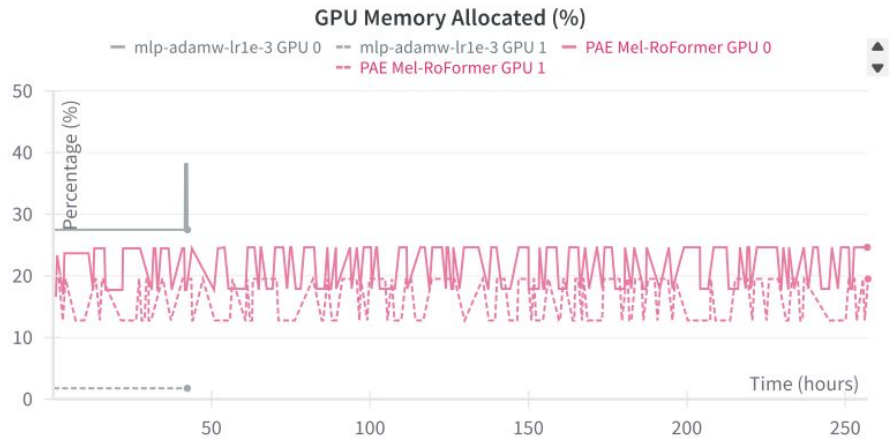



Figure 5.2: GPU memory allocation over training time for Systems C and D.

## 5.2 Subjective Listening Study Results

With the 12 participants that took part in the study, each provided multiple ratings across all combinations of systems, stimuli, and perceptual metrics, resulting in a total of 264 observations. The evaluation followed a MUSHRA-style test design, where participants rated each upmixing condition relative to others for each stimulus. Given the ordinal nature of the 7-point Likert scale used for rating, a non-parametric statistical approach was applied. The Friedman test was used to assess the main effects of the four systems across the perceptual metrics for each of the six evaluated stimuli. Post-hoc pairwise comparisons were then conducted using the Nemenyi test

to identify statistically significant differences between systems and to compute average system rankings.

### 5.2.1 Overall Ratings Across All Metrics

Table 5.1 presents the overall mean ratings aggregated across all perceptual metrics. Our proposed system (System C), highlighted in bold, performed second overall with an aggregated mean rating of 14.45. Table 5.2 summarizes the Friedman test results conducted separately for each metric. The analysis revealed a statistically significant effect of System for Overall Envelopment & Immersion ($\chi^2 = 10.816$, $p = 0.0128$), Spatial & Temporal Quality ($\chi^2 = 9.816$, $p = 0.0202$), and Spectral Quality ($\chi^2 = 10.500$, $p = 0.0148$). Figure 5.3 illustrates the distribution of listener ratings across these metrics for each processing system.

Table 5.1: Overall Combined Mean Ratings Across All Metrics

| **System** | **Total Mean Rating** | **Rank** |
|---|---|---|
| System B | 14.63 | 1 |
| **System C** | **14.45** | **2** |
| System A | 12.91 | 3 |
| System D | 12.32 | 4 |

Table 5.2: Friedman Test Across All Metrics

| **Perceptual Metric** | **Chi-square ($\chi^2$)** | **p-value** |
|---|---|---|
| Overall Envelopment & Immersion | 10.816 | **0.0128** |
| Spatial & Temporal Quality | 9.816 | **0.0202** |
| Spectral Quality | 10.500 | **0.0148** |

Post-hoc Nemenyi tests (Tables 5.3–5.5) indicated a statistically significant perceived difference between System C and System A for Overall Envelopment & Immersion ($p = 0.0181$), and between System D and both System B ($p = 0.0230$) and System C ($p = 0.0449$) for Spectral Quality. No statistically significant pairwise differences were observed for Spatial & Temporal Quality.

Table 5.3: Nemenyi Post-hoc Test for Overall Envelopment & Immersion

| | **System A** | **System B** | **System C** | **System D** |
|---|---|---|---|---|
| System A | 1.000 | 0.068 | **0.018** | 0.585 |
| System B | | 1.000 | 0.965 | 0.636 |
| System C | | | 1.000 | 0.345 |
| System D | | | | 1.000 |

Table 5.4: Nemenyi Post-hoc Test for Spatial & Temporal Quality

| | **System A** | **System B** | **System C** | **System D** |
|---|---|---|---|---|
| System A | 1.000 | 0.068 | 0.168 | 0.995 |
| System B | | 1.000 | 0.979 | 0.119 |
| System C | | | 1.000 | 0.265 |
| System D | | | | 1.000 |

Table 5.5: Nemenyi Post-hoc Test for Spectral Quality

| | **System A** | **System B** | **System C** | **System D** |
|---|---|---|---|---|
| System A | 1.000 | 0.733 | 0.859 | 0.265 |
| System B | | 1.000 | 0.995 | **0.023** |
| System C | | | 1.000 | **0.045** |
| System D | | | | 1.000 |

A closer inspection of the boxplot distributions (Figure 5.3) provides additional insight into the distribution and variance of the data. For both the Overall Envelopment & Immersion and Spatial & Temporal Quality metrics, System C and System B exhibited higher medians and upper quartiles compared to System A, despite having lower mean ratings. This suggests that a subset of listeners consistently rated these systems highly, indicating a polarized or skewed perception in those conditions. In contrast, for the Spectral Quality metric, the medians and interquartile ranges for Systems A–C were nearly identical which indicates comparable performance and greater agreement among listeners.

Across all three perceptual metrics, System D consistently received the lowest ratings, indicating a clear perceptual disadvantage. This was especially evident in Spectral Quality, although its performance in Spatial & Temporal Quality was comparable to that of System A.

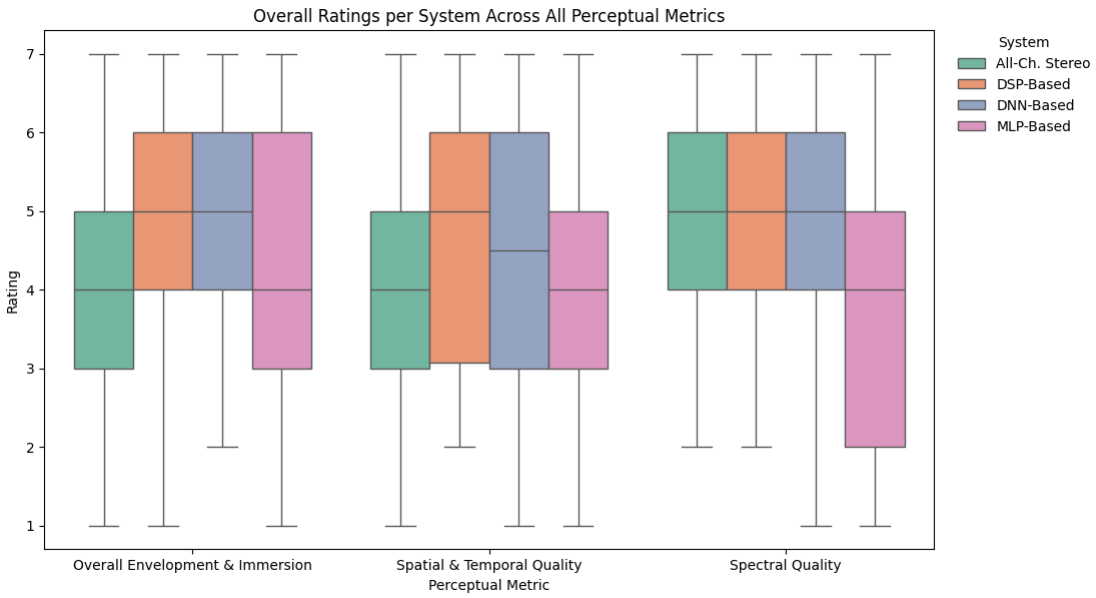



Figure 5.3: Boxplot of listener ratings for all metric ratings across upmixing systems.

### 5.2.2 Overall Envelopment and Immersion

Table 5.6 presents the average ratings for Overall Envelopment & Immersion across systems. Our proposed system (System C), highlighted in bold, was the highest ranked for this perceptual metric with a mean rating of 4.84. Table 5.7 summarizes the Friedman test results conducted per stimulus. The analysis revealed a statistically significant effect of System for Stimulus C ($\chi^2 = 9.798$, $p = 0.0204$) and Stimulus E ($\chi^2 = 19.075$, $p = 0.0003$). No statistically significant effects were observed for other stimuli.

Table 5.6: Mean Ratings for Overall Envelopment & Immersion

| **System** | **Mean Rating** | **Rank** |
|---|---|---|
| **System C** | **4.84** | **1** |
| System B | 4.77 | 2 |
| System D | 4.26 | 3 |
| System A | 3.98 | 4 |

Table 5.7: Friedman Test for Overall Envelopment & Immersion (Per Stimulus)

| **Stimulus** | **Chi-square ($\chi^2$)** | **p-value** |
|---|---|---|
| A | 3.990 | 0.2625 |
| B | 0.429 | 0.9343 |
| C | 9.798 | **0.0204** |
| D | 7.740 | 0.0517 |
| E | 19.075 | **0.0003** |
| F | 6.771 | 0.0796 |

Post-hoc Nemenyi tests (Tables 5.8–5.9) indicated a perceived difference between System C and System A for Stimulus C ($p = 0.0997$), and a statistically significant difference between System D and System C ($p = 0.0004$) for Stimulus E.

Table 5.8: Nemenyi Post-hoc Test for Stimulus C (Overall Envelopment & Immersion)

| | **System A** | **System B** | **System C** | **System D** |
|---|---|---|---|---|
| System A | 1.000 | 0.083 | **0.100** | 0.733 |
| System B | | 1.000 | 1.000 | 0.535 |
| System C | | | 1.000 | 0.585 |
| System D | | | | 1.000 |

Table 5.9: Nemenyi Post-hoc Test for Stimulus E (Overall Envelopment & Immersion)

| | **System A** | **System B** | **System C** | **System D** |
|---|---|---|---|---|
| System A | 1.000 | 0.685 | 0.055 | 0.485 |
| System B | | 1.000 | 0.485 | 0.055 |
| System C | | | 1.000 | **0.000** |
| System D | | | | 1.000 |

When comparing the distributions seen in Figure 5.4, Systems A–C appeared to perform comparably for Stimuli A and B, with small differences in median ratings and overlapping interquartile ranges. In contrast, for Stimuli C through F, System B and System C frequently outperformed System A in both median and upper quartile ratings, indicating a stronger perceived immersion for certain content types. System D showed wide variability in ratings for Stimuli A and B, and interestingly, outperformed all other systems for Stimulus D, which corresponded to the jazz audio sample.

These variations suggest that immersion perception may be influenced not only by the upmixing method but also by listener preferences and the inherent characteristics

of the source material. Factors such as genre familiarity, mixing style, and spatial content distribution likely played a role in shaping subjective evaluations across different stimuli. However, it is difficult to draw definitive conclusions from these patterns alone, and a deeper investigation into listener bias and content-specific mixing characteristics in the context of upmixing is warranted.

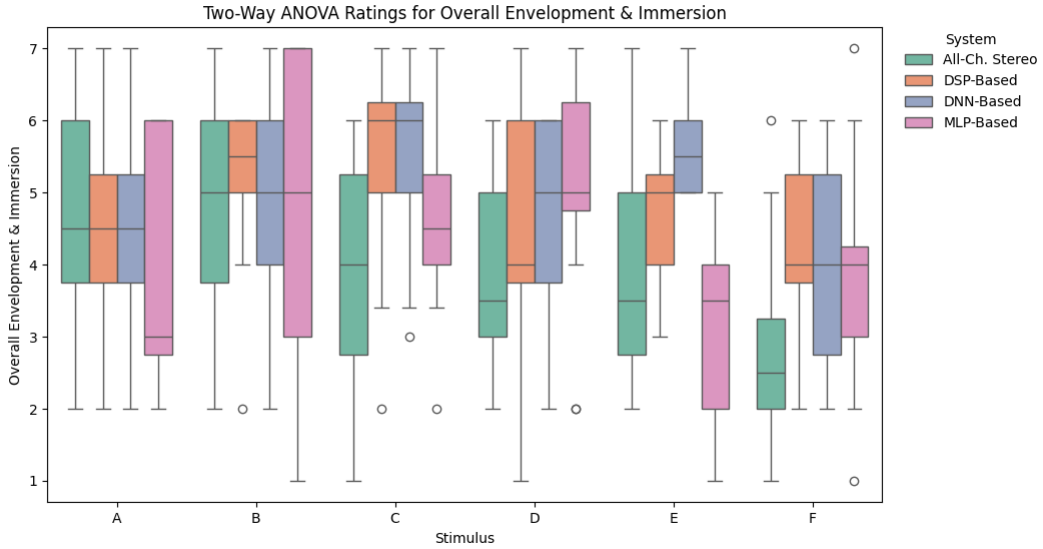



Figure 5.4: Boxplot of listener ratings for Overall Envelopment & Immersion across upmixing systems.

### 5.2.3 Spatial and Temporal Quality

Table 5.10 presents the average ratings for Spatial & Temporal Quality across systems. Our proposed system (System C), highlighted in bold, ranked second among the systems with a mean rating of 4.46. Table 5.11 summarizes the Friedman test results conducted per stimulus. The analysis once again revealed a statistically significant effect of System for Stimulus C ($\chi^2 = 13.441$, $p = 0.0038$) and Stimulus E

($\chi^2 = 19.890$, $p = 0.0002$). No statistically significant effects were observed for other stimuli.

Table 5.10: Mean Ratings for Spatial & Temporal Quality

| **System** | **Mean Rating** | **Rank** |
|---|---|---|
| System B | 4.64 | 1 |
| **System C** | **4.46** | **2** |
| System D | 4.07 | 3 |
| System A | 3.88 | 4 |

Table 5.11: Friedman Test for Spatial & Temporal Quality (Per Stimulus)

| **Stimulus** | **Chi-square ($\chi^2$)** | **p-value** |
|---|---|---|
| A | 2.301 | 0.5123 |
| B | 4.867 | 0.1818 |
| C | 13.441 | **0.0038** |
| D | 5.292 | 0.1516 |
| E | 19.890 | **0.0002** |
| F | 2.073 | 0.5574 |

Post-hoc Nemenyi tests (Tables 5.12–5.13) indicated a statistically significant perceived difference between System A and System B for Stimulus C ($p = 0.0110$), and between System A and C ($p = 0.0045$), System D and System C ($p = 0.0012$) as well as System D and System B ($p = 0.0289$) for Stimulus E.

Table 5.12: Nemenyi Post-hoc Test for Stimulus C (Spatial & Temporal Quality)

| | **System A** | **System B** | **System C** | **System D** |
|---|---|---|---|---|
| System A | 1.000 | **0.011** | 0.197 | 0.922 |
| System B | | 1.000 | 0.685 | 0.068 |
| System C | | | 1.000 | 0.535 |
| System D | | | | 1.000 |

Table 5.13: Nemenyi Post-hoc Test for Stimulus E (Spatial & Temporal Quality)

| | **System A** | **System B** | **System C** | **System D** |
|---|---|---|---|---|
| System A | 1.000 | 0.345 | **0.045** | 0.685 |
| System B | | 1.000 | 0.778 | **0.029** |
| System C | | | 1.000 | **0.001** |
| System D | | | | 1.000 |

The boxplot distributions (Figure 5.5) show that for Stimuli B through E, both System B and System C consistently outperformed System A in spatial quality, indicating an advantage in performance for the majority of music-focused audio content.

However, System A demonstrated an across-the-board advantage only in Stimulus A, which featured the most reverberant and ambient mix among the evaluated music samples. This stimulus already possessed a spacious acoustic character, and additional spatialization introduced by the upmixing systems (particularly through the additional ambient signals in the rear channels) may have disrupted the natural acoustic realism of the audio mix. This could explain the lower perceived spatial quality ratings for the upmixed versions, as the evaluation of the realism of the acoustic space is a key component of this metric. Notably, System A also performed best for Stimulus A across other metrics (Figures 5.4 and 5.6), suggesting that in certain

content cases, upmixing may not only be unnecessary but potentially detrimental to the listening experience.

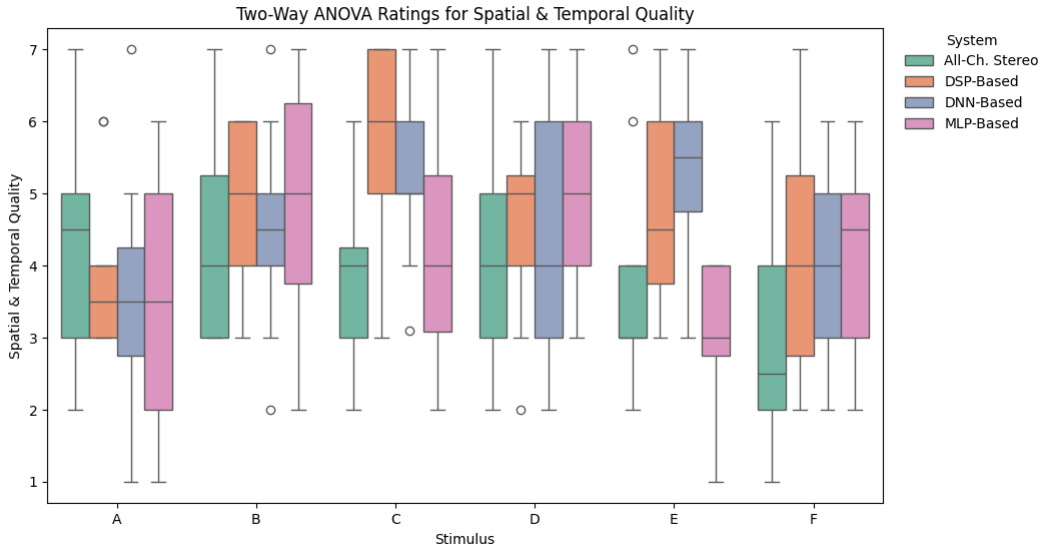



Figure 5.5: Boxplot of listener ratings for Spatial & Temporal Quality across upmixing systems.

### 5.2.4 Spectral Quality

Table 5.14 presents the average ratings for Spectral Quality across systems. Our proposed system (System C), highlighted in bold, ranked second with a mean rating of 5.15. Table 5.15 summarizes the Friedman test results conducted per stimulus. The analysis revealed a statistically significant effect of System for Stimuli A ($\chi^2 = 14.257$, $p = 0.0026$), C ($\chi^2 = 11.179$, $p = 0.0108$), and E ($\chi^2 = 16.326$, $p = 0.0010$). No statistically significant effects were observed for other stimuli.

Table 5.14: Mean Ratings for Spectral Quality

| System | Mean Rating | Rank |
|---|---|---|
| System B | 5.22 | 1 |
| **System C** | **5.15** | **2** |
| System A | 5.06 | 3 |
| System D | 3.99 | 4 |

Table 5.15: Friedman Test for Spectral Quality (Per Stimulus)

| Stimulus | Chi-square ($\chi^2$) | p-value |
|---|---|---|
| A | 14.257 | **0.0026** |
| B | 4.316 | 0.2293 |
| C | 11.179 | **0.0108** |
| D | 3.100 | 0.3765 |
| E | 16.326 | **0.0010** |
| F | 2.585 | 0.4601 |

Post-hoc Nemenyi tests (Tables 5.16–5.18) indicated a statistically significant perceived difference between System D and System A for Stimulus A ($p = 0.0028$), between System D and System C for Stimulus E ($p = 0.0028$), and between System D and System B for Stimulus E ($p = 0.0679$). No other pairwise comparisons reached significance.

This outcome aligns with the underlying design of System D's method, which generates the ambient signal through a binary classification of primary versus ambient frames throughout the audio signal. This process introduced audible artifacts in the upmixed output due to the abrupt and sometimes incorrect gating of content. These artifacts were consistently noticed and remarked upon by participants during the

listening study, as reflected in the system's lower ratings in spectral fidelity and overall tonal consistency.

Table 5.16: Nemenyi Post-hoc Test for Stimulus A (Spectral Quality)

| | **System A** | **System B** | **System C** | **System D** |
|---|---|---|---|---|
| System A | 1.000 | 0.778 | 0.389 | **0.003** |
| System B | | 1.000 | 0.922 | 0.055 |
| System C | | | 1.000 | 0.229 |
| System D | | | | 1.000 |

Table 5.17: Nemenyi Post-hoc Test for Stimulus C (Spectral Quality)

| | **System A** | **System B** | **System C** | **System D** |
|---|---|---|---|---|
| System A | 1.000 | 0.265 | 0.535 | 0.922 |
| System B | | 1.000 | 0.965 | 0.068 |
| System C | | | 1.000 | 0.197 |
| System D | | | | 1.000 |

Table 5.18: Nemenyi Post-hoc Test for Stimulus E (Spectral Quality)

| | **System A** | **System B** | **System C** | **System D** |
|---|---|---|---|---|
| System A | 1.000 | 0.859 | 0.265 | 0.345 |
| System B | | 1.000 | 0.733 | 0.068 |
| System C | | | 1.000 | **0.003** |
| System D | | | | 1.000 |

The boxplot distributions for Spectral Quality (Figure 5.6) reinforce the statistical findings in showing that System D consistently received the lowest ratings across all stimuli. This trend was visually apparent in the lower medians compared to the other

systems. Interestingly, written comments from several participants revealed a contradictory perception of System D. These anomalies, although technically degradations in spectral quality, were sometimes perceived as enhancing certain sound cues and events. This suggests that spectral artifacts may have subconsciously contributed to heightened perceptual engagement for some listeners.

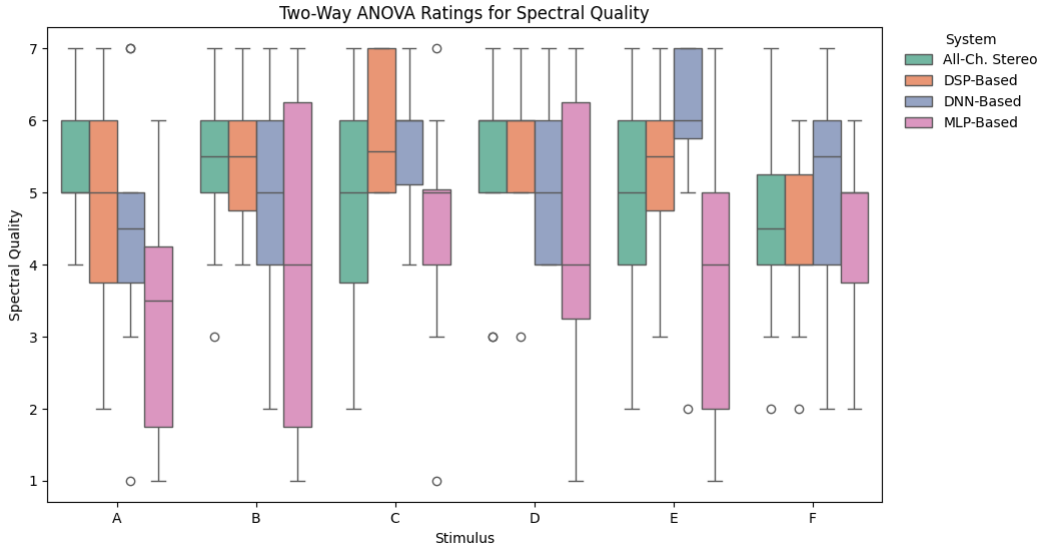



Figure 5.6: Boxplot of listener ratings for Spectral Quality across different processing systems.

## 5.3 Objective Metric Results

As outlined in Section 4.3, we employed three standard objective measures to quantitatively evaluate the upmixing performance of the trained neural networks: SDR, SI-SDR, and LSD. These metrics were computed across a test set of 12 full-length music samples, with DSP-generated ambient signals as ground truth references for comparison.

Table 5.19: Average Objective Metrics Across Methods

| System | SDR (dB) | SI-SDR (dB) | LSD (dB) |
|---|---|---|---|
| System C (Mel-RoFormer) | **2.22** | **3.44** | **106.93** |
| System D (MLP) | 0.89 | -5.29 | 446.20 |

Table 5.19 presents the average results for System C and System D. System C consistently outperformed the System D baseline across all metrics. Notably, its higher SI-SDR and lower LSD scores indicate improved signal fidelity and spectral consistency in the extracted ambient signals. In contrast, System D exhibited greater spectral deviation and lower signal-to-distortion ratios, which aligns with subjective observations of audible artifacts and signal instability.

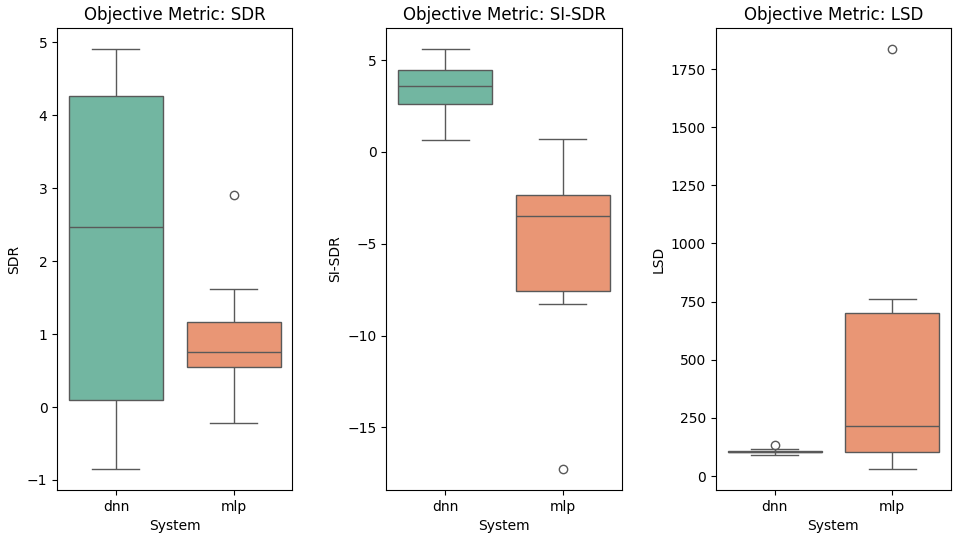



Figure 5.7: Boxplot distribution of SDR, SI-SDR, and LSD scores across test samples.

Figure 5.7 presents the boxplot distributions and highlights the variance and range of values for each metric across the test set. This visualization confirms that although SDR values are relatively close between systems, SI-SDR shows a strong separation

in favor of System C, whereas System D exhibits extremely high LSD values due to poorer spectral reconstruction.

To statistically assess these differences, one-way ANOVAs were conducted for each of the three objective metrics, as shown in Table 5.20. While the SDR metric showed a marginal difference between systems, it did not reach statistical significance. However, both SI-SDR and LSD revealed significant main effects, indicating that system choice had a substantial influence on distortion and spectral deviation performance.

Tukey's HSD post-hoc tests (Table 5.21) confirmed that System C significantly outperformed the System D baseline in both SI-SDR and LSD metrics. The large effect observed in SI-SDR highlights System C's advantage in reducing interference and reconstruction noise in the predicted ambient signal, while the LSD results reflect improved spectral preservation. These objective findings are aligned with the perceptual results reported in the listening study, reinforcing the superiority of System C's approach.

Table 5.20: One-Way ANOVA Results for Objective Metrics (SDR, SI-SDR, LSD)

| **Metric** | **Factor** | **Sum of Squares** | **df** | **F-value** | **p-value** |
|---|---|---|---|---|---|
| SDR | System | 10.63 | 1 | 3.83 | 0.063 |
| | Residual | 60.98 | 22 | - | - |
| SI-SDR | System | 457.61 | 1 | 36.58 | **0.000004** |
| | Residual | 275.20 | 22 | - | - |
| LSD | System | $6.91 \times 10^5$ | 1 | 5.09 | **0.034** |
| | Residual | $2.98 \times 10^6$ | 22 | - | - |

Table 5.21: Post-hoc Tukey's HSD for Significant Objective Metrics (SI-SDR and LSD)

| **Metric** | **group1** | **group2** | **meandiff** | **p-adj** | **lower** | **upper** |
|---|---|---|---|---|---|---|
| SI-SDR | System C | System D | -8.7332 | **0.0000** | -11.7276 | -5.7387 |
| LSD | System C | System D | 339.2701 | **0.0343** | 27.5541 | 650.9860 |

# CHAPTER 6

# Conclusion

To summarize, this thesis investigated the application of DNNs as a solution for PAE and addressing the limitations of previous DSP algorithms in the context of stereo-to-surround audio upmixing. By reframing ambient signal extraction as a source separation task, we aimed to develop a more flexible and effective approach by utilizing high-complexity MSS architectures presented in recent literature.

The previous chapters provided a comprehensive review of the SOTA systems and techniques in the fields of MSS and automatic blind upmixing, identified key knowledge gaps in the literature, and outlined the guiding research question and proposed methodologies that address the identified research gap. We detailed the design and implementation of the data procurement and preparation of the custom PAEDB dataset, the supervised learning strategy and training parameters, and the selected model architecture for PAE optimization, integrated into the extraction blocks of the VPA upmixing system. This was followed by a thorough evaluation through a designed participant listening study and objective performance metrics.

The experimental results demonstrate that the DNN-based system consistently outperformed the MLP model in both training efficiency and separation quality. The training loss curves showed that the DNN model achieved stable convergence, while

the MLP exhibited multiple plateaus and reached an earlier performance ceiling, indicating the architecture's limited capacity for learning complex ambient extraction patterns.

In terms of computational efficiency, the MLP model trained faster with lower GPU utilization, while the DNN model required significantly more time and resources despite employing data parallelization across multiple GPUs.

The subjective evaluation results revealed significant differences in listener ratings across systems and perceptual attributes. Non-parametric Friedman tests showed that the DNN-based system (System C) and the DSP-based system (System B) not only outperformed the other systems, but also closely matched each other in overall perceptual quality across stimuli. In contrast, Spectral Quality ratings highlighted a consistent disadvantage for the MLP-based system (System D), which was rated significantly lower than the others. Participant feedback attributed this to audible artifacts introduced by the MLP's gating mechanism for ambient content. Interestingly, some listeners noted that these same artifacts occasionally enhanced perceived immersion, suggesting a complex interplay between spectral fidelity and subjective engagement depending on the listening context.

The objective evaluation confirmed that SDR differences indicated a modest improvement in signal reconstruction quality, with SI-SDR showing a significant advantage for the DNN model in reducing distortion, while LSD revealed greater spectral mismatch for the MLP model, further supporting the method's limited coherence in ambient reconstruction.

Overall, these results demonstrate that the proposed DNN-based approach, specifically the Mel-RoFormer architecture trained from scratch, effectively learns the com-

plex structure of DSP-generated ambient signals. Both subjective and objective evaluations suggest that this model delivers perceptually coherent and high-quality ambient signal extraction. Notably, the Mel-RoFormer closely matches the performance of the traditional DSP algorithm (ELAE method) while maintaining the flexibility and efficiency of a learned neural architecture. Compared to the simpler MLP method proposed in [10], the more complex DNN approach provides substantial improvements in both perceived spatial quality and spectral fidelity, demonstrating the potential of deep, attention-based architectures for high-fidelity ambience extraction in the context of surround audio upmixing.

## 6.1 Discussion of Limitations

### 6.1.1 Observation of Objective Metrics and Loss Functions

Although we observed measurable improvements in the DNN model's performance across SDR, SI-SDR, and LSD, the absolute values of these metrics remain on the lower end compared to SOTA MSS models optimized for the music separation task. This suggests an inherent limitation in relying solely on conventional signal reconstruction metrics for evaluating ambient signal quality. Ambient components are, by nature, diffuse, decorrelated, and often stochastic, making them fundamentally different from structured sources like vocals or drums. To this effect, models trained to optimize over SDR or SI-SDR metrics may focus on minimizing frame-level reconstruction error rather than capturing the perceptual and spatial characteristics that define realistic ambience.

Similarly, the chosen L1 and MRSTFT losses may also struggle to guide the model toward perceptually meaningful solutions in the context of ambient extraction, as they do not explicitly account for stereo image coherence or spatial diffusion.

This mismatch between the nature of the task and the evaluation criteria may lead to suboptimal learning during training, despite achieving a satisfactory perceptual outcome. If the objective function fails to align with perceptual relevance, the network may converge on solutions that technically satisfy the loss function but fail to produce perceptually convincing results. This could explain why prior work on ambient extraction has consistently reported poor SDR values, often in the negative range [30, 31]. To put it simply, even if the generated ambient signals sound spatially diffuse and perceptually coherent, they do not necessarily align frame-for-frame with the reference signals, leading to penalization under standard objective metrics.

### 6.1.2 Data Scarcity and Subsequent Performance Ceiling

Another key limitation identified during this investigation was the scarcity of publicly available datasets for primary ambient signal extraction. While datasets for speech dereverberation and enhancement exist, they are typically constrained to speech and residual components, lacking the dry and reverberant music content required for broader music applications. As a result, we were unable to leverage large-scale, ground-truth datasets tailored specifically for ambient signal separation. To address this, we relied on DSP-based ambient extraction methods [14] to generate the training targets.

These algorithmically generated signals may inherently introduce a performance ceiling, as the model can only learn what the DSP algorithm is capable of producing.

Rather than mitigating the spectral and temporal artifacts introduced by traditional DSP methods, the model risks replicating these very artifacts. One could argue that training a DNN to mimic a DSP-based ambient extraction algorithm allows it to generalize and internalize the rule-based logic within its layers, potentially providing a strong foundation for future fine-tuning. Nonetheless, the additional computational and resource costs required for this approach must be carefully justified if the learned model continues to inherit the limitations and artifacts of the DSP algorithm it was trained to emulate.

## 6.2 Future Work

These observations underscore the need for alternative evaluation and training strategies in the context of ambient extraction. Future work should explore objective metrics that better correlate with spatial perception, such as measures of decorrelation and diffuseness to more accurately assess ambient signal quality. Without appropriate metrics, training objectives may fail to capture the nuanced perceptual attributes that listeners associate with high-quality upmixed ambience.

In parallel, it is important to move beyond standard L1 and MRSTFT losses, which primarily penalize waveform and spectral reconstruction errors, and consider custom loss functions that also better reflect the perceptual properties of ambient audio. Potential directions include losses based on mid/side processing, inter-channel correlation, or phase coherence, which could guide the model toward learning spatially rich and perceptually convincing ambient representations.

Another future directive would be the procurement and development of a dataset containing dry and wet music recordings specifically tailored for music dereverber-

ation and ambient effect extraction. Such a dataset would enable models to learn directly from real ambient signals, rather than being constrained by DSP-generated training data. This would address one of the key limitations identified in this study's evaluation, where reliance on algorithmically approximated targets introduced a performance ceiling and limited the perceptual realism of the extracted ambience.

Finally, the model architecture and feature space of current MSS models may not be inherently optimized for ambient extraction. Future research could explore the development of custom architectures explicitly designed for this task. One promising direction is the integration of latent space representations inspired by variational autoencoders (VAEs), as demonstrated in style-transfer upmixing methods [37], which have shown the ability to capture spatial instrument placement across channels. Such representations may provide a more suitable feature space for modeling the diffuse and spatially distributed nature of ambient signals.

While this study demonstrated the potential of DNN-based PAE for stereo-to-surround upmixing, several areas warrant further investigation to refine and advance the approach. Addressing these challenges may lead to more effective ambient extraction techniques, ultimately improving the perceptual quality and spatial realism of stereo-to-surround upmixing systems.

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# APPENDIX

## A Supplementary Materials

To support reproducibility and further research, the following resources are made publicly available:

### GitHub Repository

The source code is available here: `https://github.com/nick7ong/UpmixAI-Thesis`

- **Model Architectures**: PyTorch implementations of the Mel-RoFormer [9] and MLP [10] architectures.
- **Data Analysis Scripts**: Code for computing objective metrics and analyzing subjective listening test data.
- **Raw Evaluation Data**: Complete results from objective metric evaluations and subjective listening studies, including participant responses and statistical analyses.

### Hugging Face Repository

Pretrained model weights, datasets, and audio examples are hosted on Hugging Face:

- **Model Checkpoints**: Trained weights for Systems C and D discussed in Sections 3.2.3. `https://huggingface.co/nick7ong/ambience-extractors`
- **Training Dataset (PAEDB)**: The processed training dataset for Primary-Ambience Extraction separation task. `https://huggingface.co/datasets/nick7ong/paedb`
- **Test Audio Set**: A curated collection of output examples, illustrating separation and upmixing performance across various musical genres and acoustic conditions. `https://huggingface.co/datasets/nick7ong/upmix-5.1-test-set`

Source code is released under the MIT License. All resources are intended for academic use only.